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Fast-Tracking Esp Well Commissioning: A Next-Generation Model For Operational Agility And Early Oil Monetization

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Abstract

In the evolving energy industry, fast and reliable commissioning of Electric Submersible Pump (ESP) wells has become a critical lever for sustaining production, improving operational efficiency, and maximizing project cash flows. Traditional ESP commissioning workflows in Nimr field averaged between four to five days after rig-off, resulting in deferred oil production, high operational expenditures, inefficient resource utilization, and missed early revenue opportunities.

This paper presents a transformative, standardized execution model that enables one day hook-up and commissioning of ESP wells, aligning field delivery practices with the necessities of operational agility, cost competitiveness, and energy transition readiness. By systematically shifting critical activities -such as Electrical & Instrumentation (E&I) foundations, high-voltage terminations, ESP skid installations, and mechanical spool prefabrications- to the pre-rig phase, the model eliminates traditional holdups and compresses commissioning timelines without compromising operational integrity and process safety. Implementation across multiple wells demonstrated powerful value creation across key operational dimensions and value drivers:

- **Production Acceleration:** Commissioning timelines were reduced by up to 80%, allowing for the capture of over 20,800 barrels of early oil production. This early production directly enhanced Net Present Value (NPV) of the field development plan, while supporting wider organizational production targets.
- **Resource Efficiency:** Manpower and vendor mobilization were optimized by more than 50%, reducing site personnel exposure and minimizing the number of third-party interventions required post rig-off. Equipment idling and associated expenses were significantly reduced, freeing valuable resources to be invested across simultaneous projects.
- **Cost Optimization:** The standardized model substantially decreased commissioning costs by reducing redundant mobilizations, minimizing field fabrication activities, and streamlining logistics. Pre-tested skids and pre-hydrotested high-pressure spools eliminated the risks and costs associated with late-stage fabrication rework.

- **HSE Enhancements:** Reducing the total on-site work scope post rig-off considerably minimized the cumulative man-hours exposed to potential site hazards. This translated into a safer commissioning environment and supported corporate HSE performance goals.
- **Engineering and Operational Agility and Resilience:** By decoupling execution activities and pre-commissioning activities from rig move timelines, the model introduced greater flexibility into field operations. Faster tie-ins enabled early production incline, where stakeholders are better positioned to react to shifting drilling schedules or unexpected rig release variations.
- **Standardization and Future Scalability:** A structured Hook-Up Readiness Index was introduced as a proactive, field-friendly tool to assess site preparedness. This index fostered better planning discipline, earlier issue identification, and a replicable, scalable commissioning strategy that can be deployed across fields and operational environments.

Importantly, the success of the one day commissioning model extends beyond direct operational gains. By embedding principles of efficiency, early intervention, and field-readiness assessment, the model aligns with broader industry goals for sustainable energy operations. Faster production start-ups improve field economics, but they also support responsible resource stewardship which is key in maintaining credibility during the global energy transition. Moreover, the model is future ready. It lays the groundwork for seamless integration of digital tools such as automated readiness tracking, AI driven commissioning workflow predictions, and remote commissioning oversight. These innovations will further enhance efficiency, predictability, and safety in ESP deployment, contributing to the creation of resilient, low carbon oilfield development strategies.

Introduction

In the dynamic landscape of oilfield development, minimizing the time between rig release and well commissioning is increasingly vital. ESP wells, integral to sustaining and boosting production, historically faced commissioning delays attributed to complex surface infrastructure requirements and vendor dependencies.

Nimr field's commissioning process averaged 4/5 days post rig-off, exposing operations to increased cost, deferred production, and prolonged manpower allocation. This study aims to introduce a standardized commissioning model, piloted successfully in Nimr, that achieves ESP well activation within one day.

Traditional Commissioning Challenges

The primary causes of delayed ESP commissioning were identified as:

- E&I scope including foundation installation for PLC, Wi-Max, ESP skids post-rig-off.
- Extensive excavation challenges (especially hard rock zones) requiring heavy equipment mobilization.
- 48-hour vendor mobilization notifications for ESP skid and cable delivery.
- Sequential skid installation, termination, energization, and soaking.
- Complex hook-up processes involving high-pressure (Class 1500) and low-pressure (Class 300) piping, each requiring separate fabrication, inspection, and hydrotesting phases.

These activities inherently created a bottleneck, preventing harmonized commissioning operations.

To address these constraints, the model emphasizes front-loading of critical activities and integration of pre-rig construction efforts. By shifting E&I foundation work, cable laying, and skid mobilization to earlier phases, the model reduces commissioning dependency on rig-release milestones. Standardizing spool elevations and pre-fabricating high pressure spools with pre-approved designs allowed for offsite

hydrotesting, eliminating multiple sequential testing events. Additionally, extending the well pad layout facilitated parallel execution without interfering with rig operations. These refinements not only shortened the commissioning cycle but also enhanced safety, reduced site congestion, and allowed personnel to focus on value generating tasks from day one.

Beyond its immediate technical and financial benefits, the one day ESP commissioning model fosters a cultural shift within operational teams. By promoting proactive planning and cross-functional collaboration, the approach strengthens communication between drilling, construction, and commissioning teams. Regular readiness reviews and pre-rig planning workshops have become a cornerstone of this methodology, ensuring that all stakeholders are aligned on critical milestones before rig release. This collaborative environment not only minimizes last minute surprises but also builds accountability, empowering field teams to take ownership of commissioning outcomes.

Another key aspect is the focus on skill development. By introducing advanced planning tools like the Hook-Up Readiness Index, field supervisors and contractors are trained to evaluate site preparedness through measurable performance indicators. This continuous learning environment encourages innovation, where teams actively look for new ways to streamline execution without compromising safety or quality. Over time, these practices strengthen operational resilience, allowing the organization to respond quickly to changes in market conditions or unexpected operational challenges.

In essence, the one day commissioning model serves as a blueprint for future oilfield projects. It demonstrates how operational excellence, when combined with innovative thinking and structured execution, can deliver significant value while maintaining a strong commitment to safety, sustainability, and energy transition goals.

Statement of Theory and Definitions

The objective of this paper is to introduce and validate a transformative execution model designed to enable one day commissioning of Electric Submersible Pump (ESP) wells in the Nimr field. This model addresses longstanding challenges associated with deferred production, high commissioning costs, and inefficient resource allocation following rig release. By integrating pre-rig construction strategies and process standardization, the model aims to:

- Accelerate early oil by reducing ESP commissioning time from 4–5 days to just one day.
- Optimize manpower and contractor mobilization to reduce exposure and overhead costs.
- Establish a scalable commissioning framework applicable across various well types and geographies.
- Enhance HSE performance by minimizing on-site work post rig-off.
- Introduce a proactive readiness assessment tool, the Hook-Up Readiness Index, to standardize field readiness and support continuous improvement.

This initiative not only streamlines execution but also redefines the role of commissioning within the overall well delivery lifecycle. By shifting from a fragmented, post-rig approach to a front-loaded, integrated strategy, the model ensures alignment between engineering, construction, and operations teams from the earliest stages. This early collaboration leads to fewer surprises, faster decision-making, and greater schedule certainty. Furthermore, the model lays the foundation for embedding digital enablement tools - such as progress tracking dashboards and remote validation systems- into the commissioning workflow. These advancements support PDO's broader vision for agile, high-performance field development in an increasingly competitive energy landscape.

Description and Application of Equipment and Processes

The new model was developed through a structured analysis of traditional ESP commissioning workflows, followed by a strategic reengineering of critical path activities. The foundation of the method involves shifting key tasks to the pre-rig phase. These include:

- Installation of Electrical & Instrumentation (E&I) foundations
- Delivery and placement of ESP skids
- Pre-laying and termination of high voltage cables
- Prefabrication and pre-hydrotesting of high pressure spools

The execution framework introduced a Hook-Up Readiness Index (HURI), a 5 point assessment tool used by field teams to verify site readiness for one day commissioning. The index ensures that all prerequisites are completed before rig release. In addition, the process emphasizes cross functional alignment and early stakeholder engagement. Weekly planning huddles, joint site walks, and interface clarifications with drilling, construction, and commissioning teams were established. Standard Operating Procedures (SOPs) and field instructions were revised to reflect the new execution strategy, supported by enhanced contractor onboarding sessions and performance dashboards for real-time monitoring.

Detailed Methodology

To achieve one day commissioning of ESP wells post rig off, altering the construction sequence of activities as follows is critical:

Electrical and Instrumentation (E&I). Current Practice:

- Foundations for PLC, Wi-Max, and ESP skid are installed post-rig-move due to footprint constraints.
- The following challenges contribute to delays:
 1. Hard rock excavation requiring mobilization of rock breakers, taking over 1.5 days.
 2. 48-hour notification required for ESP vendors to mobilize the ESP skid and surface cable.
 3. ESP vendor requiring half a day for cable termination and configuration.
 4. Soaking hours for skid after energization. (minimum of 4 hours)

Proposed Solution:

- Lay skid foundation, surface cable, and instrument cable prior to rig-on.
- Extend the well pad (10Mx20M) to accommodate the skid installation before rig-on.
- Align with the rig team to ensure drilling operations proceed smoothly without impacting the skid.
- Complete PLC and UDB panel installation and termination prior to rig-on.
- Conduct HV termination and testing prior to rig-on.
- Perform skid testing before rig-on.

Hook-Up Arrangement. Current Challenges:

- ESP hook-up piping consists of two major piping classes (#300/#1500), requiring two different hydro-test pressures (6645/19940 KPa).
- High-pressure spools must be fabricated, NDT inspected, and hydro-tested at the yard, taking three days post-handover.

- Low-pressure hook-up piping must be hydro-tested separately on-site.

Proposed Solution:

- Standardize the wing valve elevation at 1050 mm above ground level, allowing for pre-fabrication and hydro-testing of the high-pressure spool before rig-on.
- Coordinate with the rig team on BOP size for upcoming wells to ensure pre-fabrication of the high-pressure spool without the need for modifications.
- Modify one spool in the low-pressure piping (highlighted in below) by converting it from ROTO to bare CS spool and increasing its schedule from SCH40 to SCH120/160. This will help manage elevation differences post-rig-off, reducing the need for high-pressure spool modifications.
- Adjust high-pressure spool sleepers as necessary to ensure proper pipe resting.

Presentation of Data and Results

The deployment of the one day commissioning model across multiple ESP wells in the Nimr field has demonstrated considerable operational gains:

- **Production Acceleration:** The average time from rig-off to production start was reduced by over 80%, enabling the capture of more than 20,800 barrels of early oil. This directly improved Net Present Value (NPV) of field developments.
- **Resource Optimization:** Vendor and crew mobilizations were reduced by 50% on activity post the rig release, minimizing idle time, site congestion, and HSE exposure.
- **Cost Reduction:** Cost savings were realized through reduced field fabrication, fewer third-party interfaces, and optimized logistics. Prefabricated, pre-tested equipment eliminated rework and redundant site tasks.
- **HSE Improvements:** With fewer on-site activities after rig-off, overall man-hour exposure was significantly reduced, supporting PDO's corporate safety objectives.
- **Execution Flexibility:** The decoupling of rig move timelines and surface commissioning activities introduced scheduling agility, improving operational resilience.
- **Replicability:** The Hook-Up Readiness Index ensured consistent field execution and became a central KPI to drive discipline across teams.

This initiative was validated through internal reviews and feedback, establishing it as a commissioning best practice that is suited for replication in other asset clusters.

Implementation and Root Cause Analysis

The implementation of the one day ESP commissioning model was guided by structured Lean principles and field-proven construction management tools. A phased deployment strategy was adopted, beginning with pilot wells to validate assumptions and workflows before scaling to a broader field base.

Root cause analysis was conducted using process mapping, fishbone diagrams, and the 5 Whys technique. Key bottlenecks in the traditional workflow were identified as:

- Delayed E&I foundation installation post rig-off
- Last-minute delivery of ESP skids, often triggering rework
- Cable laying dependent on rig movement and trench availability
- On-site fabrication of spools, prone to dimensional mismatches and delays
- Back-to-back Hydrotest of #1500 and #300 spools, due to spool fabrication requirement after rig release.

The root causes were primarily linked to fragmented scheduling, limited pre-rig engagement, and the absence of a unified readiness checklist.

To mitigate these, the following corrective actions were taken:

- Pre-fabrication of spools and change #300 spool to be ready to adjustment, which lead to hydrotest #1500 spool in yard, leaving only one hydrotest required after rig-off.
- Delivery of skids and materials at least one week before rig-off
- Realignment of project Gantt charts to include pre-rig commissioning steps
- Development of a readiness tracking system tied to the Hook-Up Readiness Index

These actions were institutionalized across planning cycles, with regular field audits and coaching sessions to embed the new practices into contractor behavior.

Table 1—Five whys that help creating One day commissioning HU Readiness Checklist

#	Why?	Root Cause
1	Why is the commissioning delayed?	Because key E&I and mechanical works start only after rig-off.
2	Why do these works start post rig-off?	Because foundations, skids, cables, and spools are not installed or prefabricated in advance.
3	Why are these items not prepared in advance?	Because current construction sequencing depends on rig movement and post-handover execution.
4	Why is the construction sequence dependent on rig movement?	There is no integrated pre-rig construction strategy or readiness trigger in place.
5	Why is there no readiness trigger?	Because the team lacks a standardized early-readiness planning

Table 2—Five whys that helped identify that #300 spool can be modified when it is changed to bare spool and higher schedule

#	Why?	Root Cause
1	Why is the commissioning delayed?	Because we need to hydrotest the fabricated #1500 spool after missing the difference in elevation then to hydrotest the HU including #300 spools.
2	Why you need to fabricate #1500 spool ?	Because you can't fabricate #300 spool instead of #1500
3	Why can't you fabricate #300 spool?	Because its PE-coated spool and couldn't be modified easily after rig-off.
4	Why couldn't the spool be modified on-site?	Because PE coating complicates welding and makes late-stage rework slow and risky.
5	Why is rework so sensitive for these spools?	Because it can not be modified with being bare with higher schedule.

Business Outcome

The business value generated by this new commissioning approach has been significant. Within the first set of pilot implementations, the following measurable outcomes were recorded:

- Oil Production: Over 20,800 barrels of oil were captured early, directly contributing to improved project NPV and faster cash flow realization.
- Cost Avoidance: Field execution costs were reduced by eliminating multiple vendor mobilizations and minimizing site working hours.
- Manpower Optimization: Contractor teams were rotated more efficiently, reducing standby costs and maximizing productivity.
- Cycle Time Compression: Time from rig-off to stable production was consistently achieved within 24 hours.
- Improved Scheduling Certainty: The new model provided a more reliable commissioning window, reducing downstream impacts on operations and logistics.

The approach also supported performance driven planning by enabling field teams to better forecast material needs, contractor availability, and commissioning readiness across multiple assets.

Sustainability

To ensure sustainability, the commissioning model was embedded into Nimr cluster's standard workflows and asset planning cycles. The Hook-Up Readiness Index was formalized as a mandatory deliverable prior to rig-off for all ESP wells within the Nimr field.

A series of institutional mechanisms were introduced to support ongoing compliance:

- Updated construction readiness checklists
- Performance dashboards linked to commissioning timelines
- Monthly readiness audits
- Field coaching and knowledge-sharing sessions for new supervisors and contractors

Furthermore, the model has been positioned as a benchmark for replication in other directorates. Regular cross-functional reviews and lessons-learned sessions are to be held to refine the process, incorporate feedback, and improve planning precision. The vision extends toward a fully digitized readiness model integrated with PDO's operational systems, allowing predictive analytics and remote validation to enhance speed, safety, and reliability of ESP well commissioning into the future.

Project Impact and Benefits

The adoption of the one day ESP commissioning model has delivered transformative impact across project performance, operational culture, and stakeholder alignment.

- Strategic Oil Monetization: Early oil production enabled by the model significantly accelerated cash flow, aligning with corporate goals for value creation.
- Standardization Across Clusters: The execution approach was arranged into a scalable framework, facilitating cross-asset deployment.
- Capacity Building: Field personnel gained hands-on exposure to structured commissioning practices, improving overall team maturity and capability.
- Contractor Integration: Clear expectations and readiness metrics improved alignment with vendors, reducing ambiguity and enhancing collaboration.
- Cross-Functional Synergy: The initiative bridged planning, engineering, construction, and operations, fostering a performance culture driven by results.

These impacts are not only tactical but strategic, positioning PDO's well delivery and commissioning teams to lead the shift toward more agile, lean, and predictable project models across the organization.

Risk Management

As with any accelerated operational model, risk identification and mitigation were central to the design and commencement of the one day ESP commissioning strategy.

Risk categories and corresponding mitigation strategies included:

- HSE Exposure: Pre-rig activities were planned with full compliance to PDO's HEMP (Hazards and Effects Management Process) framework. Modularization reduced field exposure and SIMOPS risk.
- Quality Assurance: All prefabricated items were verified against design tolerances.

- **Schedule Slippage:** Readiness reviews and gating mechanisms were introduced to ensure that no well entered commissioning without a score of 5/5 on the Hook-Up Readiness Index.
- **Interface Gaps:** Multi-discipline planning huddles and stakeholder alignment sessions were held weekly to anticipate miscommunication and interface delays.

No incidents were recorded during the pilot phase. Lessons learned were captured and translated into updated procedures, further reinforcing process robustness.

Quality of Project Management

The success of the commissioning model was underpinned by a well-organized application of project management principles. Key enablers of quality included:

- **Strong Stakeholder Alliance:** The collaboration of each stakeholder towards achieving the one day commissioning is vital to enable the model success.
- **Integrated Planning:** Master schedules incorporated pre-rig commissioning milestones and linked these to procurement, construction, and drilling plans.
- **Performance Tracking:** Visual management tools and digital dashboards provided real-time visibility into commissioning readiness and performance.
- **Stakeholder Accountability:** Defined RACI matrices ensured clear ownership of tasks, with regular progress reviews and risk logs maintained.
- **Process Discipline:** The team adopted PDCA (Plan-Do-Check-Act) cycles to continuously refine execution practices and deliver measurable improvements.

This disciplined and transparent approach ensured not only the initial success of the model, but its ongoing evolution and institutional adoption within PDO's broader well delivery strategy.

Novel/Additive Information

The novelty of this approach lies in its structured transformation of commissioning workflows into a pre-rig enabled execution model. Key innovative elements include:

- **Hook-Up Readiness Index (HURI):** A field-applicable, quantifiable index that measures commissioning readiness across five critical inputs. It provides a shared framework for planners, engineers, and field teams to align expectations and execution pace.
- **Pre-Rig Execution Strategy:** Shifting high voltage terminations, cable laying, skid delivery, and spool testing to the pre-rig phase drastically improved lead times.
- **Field-Scalable SOPs:** Standard operating procedures and pre-checklists were revised to fit various ESP well configurations.
- **Digital Readiness:** The model lays a foundation for integration with digital commissioning tools, including automated tracking systems and AI based predictive scheduling.
- **Early Monetization Model:** By emphasizing early production incline, the model introduces a financial lens to engineering execution, improving both project economics and delivery speed.

Table 3—ESP Hook-Up Readiness Index, Total score of 5/5 = Fully ready for one day commissioning.

Item	Status	Score
Foundation installed	Yes	1
ESP Skid delivered	Yes	1
Cables pre-laid and terminated	Yes	1
Pre-hydrotested high-pressure spools	Yes	1
PLC/UDB panels tested	Yes	1

These elements represent a model shift in how commissioning is abstracted, enabling oilfield operators to approach ESP deployment not as a final step, but as a strategic lever for value realization. By systematizing these innovations, the model ensures consistency across deployments, reducing variability that often leads to delays and cost overruns. The Hook-Up Readiness Index, in particular, fosters a culture of accountability and forward planning, encouraging site teams to engage in early issue identification and resolution. Moreover, embedding commissioning requirements into the front end of the project timeline enables better vendor coordination, streamlined logistics, and minimized waiting time between activities. This pre-emptive methodology empowers teams to anticipate rather than react, enhancing agility in dynamic field conditions. Collectively, these innovations offer a replicable solution for future ready oilfield development strategies.

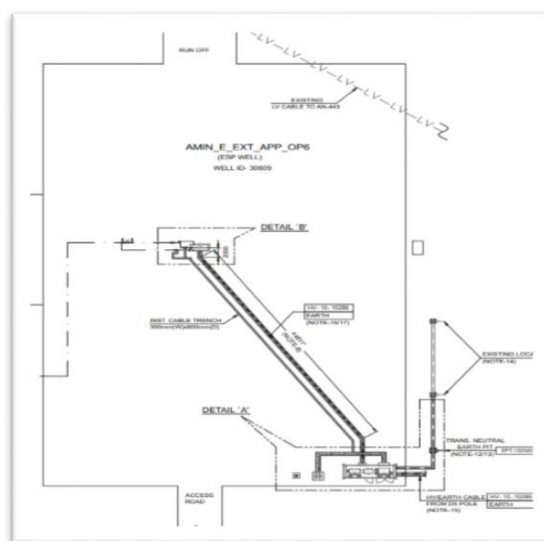


Figure 1—ESP Well Construction Drawings

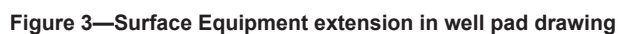




Figure 5—ESP Well H/U, #1500 & #300 Spools

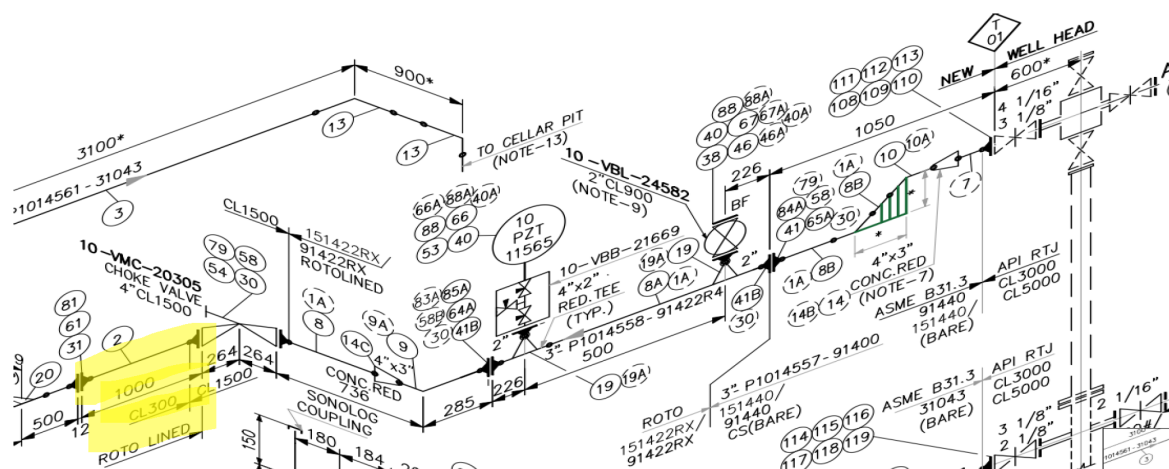


Figure 6—Spool modified from ROTO to bare CS spool

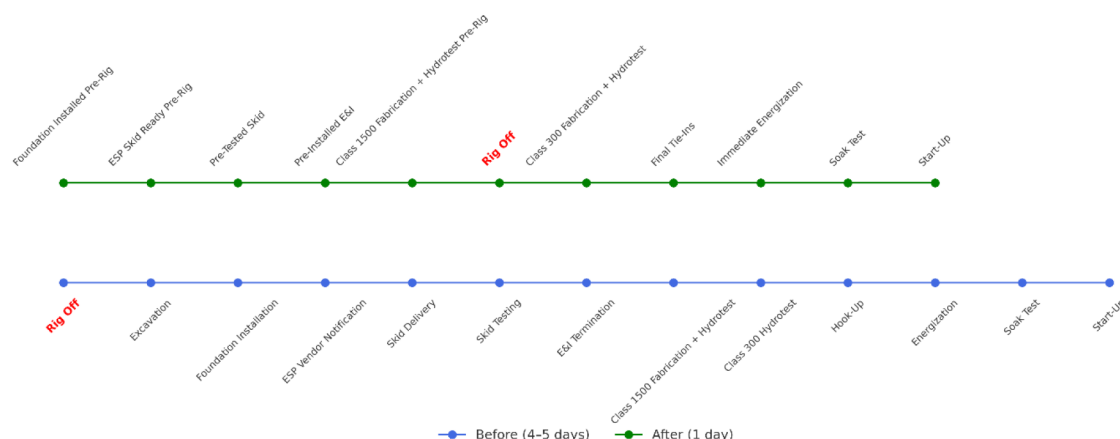


Figure 7—ESP road maps between 4/5 days and 1 day commissioning

Conclusions

The implementation of a one day commissioning model for ESP wells marks a significant milestone in the evolution of oilfield operations at PDO. By proactively addressing traditional inefficiencies and re-engineering the sequence of commissioning activities, the model has demonstrated tangible benefits

in production acceleration, cost optimization, and HSE performance. It has shifted the paradigm of commissioning from a reactive, post-rig activity to a strategic, pre-planned, value adding process.

The introduction of the Hook-Up Readiness Index provided the necessary structure to drive consistency, enforce discipline, and enable scalability. The results, faster oil monetization, streamlined logistics, and enhanced field resilience, highlight the model's potential for broader application across PDO and the industry. Ultimately, this initiative exemplifies how lean thinking, early planning, and cross-functional collaboration can transform even routine activities like ESP commissioning into critical enablers of project and business success. The model lays a strong foundation for future integration with digital tools and AI powered operational planning, ensuring that commissioning excellence continues to evolve in step with the demands of a fast changing energy landscape.

Key Takeaways:

- Accelerated Production Start-Up: Enabled capture of significant early production volumes, enhancing project cash flows and strategic field performance.
- Optimized Resource Utilization: Streamlined workforce deployment, reduced vendor dependency, and minimized equipment idling.
- Lowered Project Costs: Achieved measurable cost savings by compressing commissioning timelines and reducing on-site operational complexity.
- Enhanced HSE Outcomes: Reduced total man-hours exposed to field hazards through pre-emptive commissioning strategies.
- Built-in Flexibility and Scalability: Developed a field tested, scalable commissioning framework adaptable to various operational contexts.
- Strategic Alignment with Energy Transition Goals: Promoted operational agility, digital integration readiness, and sustainable resource management practices.

By reframing commissioning workflows and embedding operational excellence into the DNA of ESP deployment, this model transforms well delivery from a routine task into a strategic enabler of both immediate financial success and long-term industry resilience. Furthermore, the long-term impact of this model extends beyond immediate operational metrics. It establishes a cultural shift across well delivery teams, instilling a mindset that views commissioning not as an endpoint, but as a strategically orchestrated phase of the project lifecycle. This shift empowers front-line supervisors, engineers, and contractors to engage more actively in planning and decision-making, fostering a sense of ownership and accountability that enhances execution quality. It also enables PDO to respond more effectively to operational uncertainties, such as unplanned rig movement or changes in drilling sequence, through modular and adaptable workflows. From a business standpoint, the model also improves asset utilization. By reducing idle time for rigs, vendors, and support services, the initiative contributes to higher return on capital employed (ROCE) across development clusters. Additionally, the introduction of pre-commissioning metrics and readiness scores enables data driven performance reviews and continuous improvement cycles, unlocking new benchmarking opportunities for future well campaigns. As the industry continues to prioritize low carbon, high efficiency development models, PDO's one day ESP commissioning strategy positions it at the forefront of this evolution. By embedding agility, sustainability, and technology readiness at the heart of execution, the model delivers not just wells but smarter, faster, and more resilient oilfield operations.

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Nomenclature

CI	- Continuous Improvement
E&I	- Electrical and Instrumentation
ESP	- Electric Submersible Pump
FLAF	- Flowline Authorization Form
HEMP	- Hazards and Effects Management Process
HSE	- Health, Safety, and Environment
HURI	- Hook-Up Readiness Index
KPI	- Key Performance Indicator
NPV	- Net Present Value
PDO	- Petroleum Development Oman
SOP	- Standard Operating Procedure
SIMOPS	- Simultaneous Operations
VMB	- Visual Management Board